



ADVERTISEMENT

The site of the March 22 Oso mudslide shows the extent of the destruction.
PHOTOGRAPH BY ELLEN M. BANNER, THE SEATTLE TIMES VIA AP

Causes of Deadly Washington Mudslide Revealed in Scientific Report

The report says that logging in the area may have played a role.

BY **WARREN CORNWALL**, FOR NATIONAL GEOGRAPHIC



PUBLISHED JULY 24, 2014 • 5 MIN READ

A deadly landslide that killed 44 and obliterated a riverside neighborhood in Washington state last March was fueled by rain soaking the site of an eight-year-old landslide, while logging in the area may have also played a role, according to a [scientific report released Tuesday](#).

The most detailed published scientific account of the mudslide, which was the deadliest in U.S. history, suggests that the disaster was years in the making in a valley with a history of huge landslides dating back thousands of years.

"That landslide mass prior to March 22 was really in some sense poised to fail. It was really unstable," says [Joseph Wartman](#), a University of Washington engineer and a lead author of the report.

The report doesn't offer a definitive explanation for why the mountainside collapsed on that day. But it describes a devastating chain reaction sparked

by rain and groundwater on a hillside left unsettled by years of smaller slides.

The report comes from a team of university and private-sector researchers who are part of the [Geotechnical Extreme Events Reconnaissance \(GEER\)](#), a National Science Foundation-funded initiative to quickly dispatch scientists to evaluate natural disasters.

The slide plowed through part of the town of Oso, in the Cascade Mountain foothills northwest of Seattle, shortly after 10:30 a.m. on Saturday, March 22. It killed 43 people and demolished sections of State Highway 530.

The new report raises questions about how local governments take into account risks of this sort, and how the newest, best science finds its way to the local level. The hillside that collapsed had been the scene of multiple smaller slides since the first housing development was built there in the 1960s. Government agencies repeatedly attempted to buttress the hillside, though engineers warned it might do little good.

At the time of the slide, scientists were shocked at its speed and the distance it traveled—more than half a mile (over a kilometer). Yet recent images of the valley taken using LIDAR, a laser-based technology that can "see" through vegetation to the rock below, revealed multiple ancient landslides, including one more massive than the Oso slide. The GEER team also found the landslide's magnitude wasn't unusual for slides of this type.

ADVERTISEMENT

"I don't think it was fully comprehended how far out that landslide could have run. So I don't think that even local officials understood the gravity of that," Wartman says.

The incident also underscores the importance of focusing on the potential damage of a natural disaster when government officials make decisions, the scientists say. Historically, governments have tried to assess such hazards but haven't always focused on places where the consequences would be most dire.

"The very first thing is to communicate very clearly what the risks are," says [Robert Gilbert](#), a civil engineer at the University of Texas in Austin. "Not saying at one meeting every decade that 'Hey, there's a risk of landslides.' But constantly and persistently communicating what the risks are so that people are making decisions in an informed way."

The study could become fodder for legal claims already filed by survivors and families of some of the victims against Snohomish County and the state of Washington. Those focus on the state's role in regulating logging near the slide area, and the county's part in permitting homes to be built there. The county recently imposed a six-month ban on construction in the area.

The new report found that a 2006 landslide likely set the stage for the much bigger slide, creating a loose bed of rocks and soil that soaked up water more readily and was more prone to failing. Then heavy rains in the weeks leading up to the disaster drenched the region.

The slide was really two interconnected events, the report says. First, the 2006 landslide debris liquefied, a phenomenon in which solid earth becomes a liquid as the water pressure in the soil increases, pushing dirt particles apart. That then sent a mass of mud and debris shooting across the valley, removing support for a large chunk of the mountain, which fell away and added to the slide.

In the tragedy's aftermath, suspicion fell on recent logging on the mountain just above the slide and erosion along the Stillaguamish River, which flows along the base of the slide.

ADVERTISEMENT

The new report suggests there was no single cause. "Our work did not conclude that only one path could have led to the Stage 1 failure," it says, referring to the first stage of the slide.

While the report doesn't definitively point a finger at the timber industry, it suggests that logging above the slide area might have changed the way rain soaked into the hillside, adding more water to the unstable slope.

"It is possible that in 2014, the location, size and maturity of growth in these [logging] tracts was such that groundwater discharge to the slope was greater in 2014," the report states.

Getting a clearer picture of logging's role would take more intensive analysis and modeling of the groundwater flow and forestry practices there, said David Montgomery, a University of Washington geomorphologist and member of the GEER team.

The GEER scientists largely discounted river erosion as a major factor in the current slide. Someone had inspected the area just days before the slide and reported seeing little evidence of erosion along a log wall built to shield the hillside from the river, Wartman says.

But it's possible that even a little erosion could provide a final nudge on a hill already poised to collapse, Montgomery says.

Follow Warren Cornwall on [Twitter](#).

Updated 7/21/14 at 6:45 p.m.

 SHARE

 TWEET

 EMAIL

READ THIS NEXT



ENVIRONMENT

Japan poised to release nuclear wastewater into t...

The plan to gradually discharge more than a million tons of treated water from the crippled Fukushima nuclear plant has deeply divided nations and scientists.



HISTORY & CULTURE

'World's worst shipwreck' was bloodier than we thought



ANIMALS

World's first ultrasounds of wild manta rays reveal a troubling truth